

Hamiltonian

I am considering the Hamiltonian of a system of N particles with positions $\{\mathbf{R}_l\}_{l \in \{1, \dots, N\}}$. The Hamiltonian shall consist of two-body interactions, which depend on the position of the involved particles. Thus the Hamiltonian can be written in the form

$$H(\{\mathbf{R}_l\}) = \sum_{i,j; i \neq j} V_{i,j}(\mathbf{R}_i, \mathbf{R}_j)$$

In order to being able to derive the second order expansion of the Hamiltonian, that will be needed, in a more formal way, I write the two-body interactions as dependend on all positions.

$$H(\{\mathbf{R}_l\}) = \sum_{i,j; i \neq j} V_{i,j}(\{\mathbf{R}_l\})$$

The $V_{i,j}$ are formally functions of all positions $\{\mathbf{R}_l\}$, i.e. $V_{i,j}(\{\mathbf{R}_l\})$. But of course they only depend on $\mathbf{R}_i$ and $\mathbf{R}_j$, i.e.

$$\frac{\partial V_{i,j}}{\partial \mathbf{R}_l} = 0 \quad \text{if } i \neq l \neq j$$

Now I expand the Hamiltonian in small displacements from the tweezer centers $\{\delta \mathbf{R}_l\}$. By redefining the trap center the first order term can be absorbed. Consider a function $f(r)$ which is expanded up to first order in r , i.e. $f(r + \delta r) \approx f(r) + \partial_r f(r) \cdot \delta r$, if we now add a trap-like potential

$$\begin{aligned} \frac{1}{2} m \omega^2 \delta r^2 + f(r) + \partial_r f(r) \delta r &= \frac{1}{2} m \omega^2 \left(\delta r + \frac{\partial_r f(r)}{m \omega^2} \right)^2 + \frac{1}{2} \frac{(\partial_r f(r))^2}{m \omega^2} + f(r) \\ &=: \frac{1}{2} m \omega^2 \delta \tilde{r}^2 + \text{const.}, \end{aligned}$$

Thus by discarding the first order the Hamiltonian up to second order reads

$$\begin{aligned} H(\{\mathbf{R}_l + \delta \mathbf{R}_l\}) &= H(\{\mathbf{R}_l\}) + \frac{1}{2} \sum_{n,m} \delta \mathbf{R}_n \frac{\partial^2 H(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \delta \mathbf{R}_m \\ &= H(\{\mathbf{R}_l\}) + \frac{1}{2} \sum_{n \neq m} \delta \mathbf{R}_n \frac{\partial^2 H(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \delta \mathbf{R}_m + \frac{1}{2} \sum_{n=m} \delta \mathbf{R}_n \frac{\partial^2 H(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \delta \mathbf{R}_m \\ &= H(\{\mathbf{R}_l\}) + \frac{1}{2} \sum_{n \neq m} \sum_{i,j; i \neq j} \delta \mathbf{R}_n \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \delta \mathbf{R}_m + \frac{1}{2} \sum_n \sum_{i,j; i \neq j} \delta \mathbf{R}_n \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n^2} \delta \mathbf{R}_n \\ &= H(\{\mathbf{R}_l\}) + \frac{1}{2} \sum_{n \neq m} \sum_{i,j; i \neq j} \delta \mathbf{R}_n \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m} \delta \mathbf{R}_m (\delta_{n=i, m=j} + \delta_{n=j, m=i}) \\ &\quad + \frac{1}{2} \sum_n \sum_{i,j; i \neq j} \delta \mathbf{R}_n \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n^2} \delta \mathbf{R}_n (\delta_{n=i} + \delta_{n=j}) \end{aligned}$$

Using the fact, that

$$\delta \mathbf{R}_i \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i \partial \mathbf{R}_j} \delta \mathbf{R}_j = \delta \mathbf{R}_j \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_j \partial \mathbf{R}_i} \delta \mathbf{R}_i$$

the expansion can be written as

$$\begin{aligned} H(\{\mathbf{R}_l + \delta \mathbf{R}_l\}) &= H(\{\mathbf{R}_l\}) + \sum_{i,j; i \neq j} \delta \mathbf{R}_i \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i \partial \mathbf{R}_j} \delta \mathbf{R}_j \\ &\quad + \sum_{i,j; i \neq j} \delta \mathbf{R}_i \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_i^2} \delta \mathbf{R}_i \end{aligned}$$

If I introduce a matrix U the expanded Hamiltonian can be brought into the same form as in the paper.

$$H(\{\mathbf{R}_l + \delta\mathbf{R}_l\}) = H(\{\mathbf{R}_l\}) + \sum_{n,m} \delta\mathbf{R}_n U_{n,m} \delta\mathbf{R}_m$$

with

$$U_{n,m} = \frac{\partial^2 V_{n,m}(\{\mathbf{R}_l\})}{\partial\mathbf{R}_n \partial\mathbf{R}_m} \quad \text{if } n \neq m$$

$$U_{n,n} = \sum_{j; j \neq n} \frac{\partial^2 V_{n,j}(\{\mathbf{R}_l\})}{\partial\mathbf{R}_n^2}$$

Swithing again to the reduced writting, where V_{ij} is only written as a function of the variables of which it actually depends, reads

$$U_{n,m} = \frac{\partial^2 V_{n,m}(\mathbf{R}_n, \mathbf{R}_m)}{\partial\mathbf{R}_n \partial\mathbf{R}_m} \quad \text{if } n \neq m$$

$$U_{n,n} = \sum_{j; j \neq n} \frac{\partial^2 V_{n,j}(\mathbf{R}_n, \mathbf{R}_j)}{\partial\mathbf{R}_n^2}$$

This result is a good point to specify the potential that is used in the next sections further. The potential will only depend on the relative vector between the two positions, i.e. $\mathbf{r}_{ij} = \mathbf{R}_i - \mathbf{R}_j$. Using the derivative

$$\frac{\partial\mathbf{r}_{ij}}{\partial\mathbf{R}_i} = \mathbb{1} = -\frac{\partial\mathbf{r}_{ij}}{\partial\mathbf{R}_j}$$

the derivatives of the U matrix can be written as

$$\begin{aligned} \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial\mathbf{R}_n \partial\mathbf{R}_m} &= \frac{\partial}{\partial\mathbf{R}_n} \left(\frac{\partial V_{n,m}(\mathbf{r}_{nm})}{\partial\mathbf{r}_{nm}} \frac{\partial\mathbf{r}_{nm}}{\partial\mathbf{R}_m} \right) \\ &= \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial\mathbf{r}_{nm}^2} \frac{\partial\mathbf{r}_{nm}}{\partial\mathbf{R}_m} \frac{\partial\mathbf{r}_{nm}}{\partial\mathbf{R}_n} + \frac{\partial V_{n,m}(\mathbf{r}_{nm})}{\partial\mathbf{r}_{nm}} \frac{\partial^2 \mathbf{r}_{nm}}{\partial\mathbf{R}_n \partial\mathbf{R}_m} \\ &= -\frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial\mathbf{r}_{nm}^2} \end{aligned}$$

and

$$\begin{aligned} \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial\mathbf{R}_m^2} &= \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial\mathbf{r}_{nm}^2} \frac{\partial\mathbf{r}_{nm}}{\partial\mathbf{R}_m} \frac{\partial\mathbf{r}_{nm}}{\partial\mathbf{R}_m} \\ &= \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial\mathbf{r}_{nm}^2} \end{aligned}$$

Thus

$$U_{n,m} = -\frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial\mathbf{r}_{nm}^2} \quad \text{if } n \neq m$$

$$U_{n,n} = \sum_{j; j \neq n} \frac{\partial^2 V_{n,j}(\mathbf{r}_{nj})}{\partial\mathbf{r}_{nj}^2}$$

Aligned Dipole-Dipole Interactions

In this section the derivative of the dipole interaction will be performed in the case where all dipoles point into the same direction. The subsripts labeling the different positions are dropped in order to make the notation plainer.

$$V_{dd}(\mathbf{r}) = \frac{1}{r^3} - 3 \frac{(\mathbf{r} \cdot \hat{\mathbf{d}})^2}{r^5},$$

with $\mathbf{r} = \mathbf{R}_i - \mathbf{R}_j$ beeing the distance between any to sites i and j ($i \neq j$). The first derivative with respect to component α of $\mathbf{r}$, $\alpha \in \{x, y, z\}$ reads

$$\frac{\partial V_{dd}}{\partial r_\alpha} = -\frac{3r_\alpha}{r^5} + \frac{15r_\alpha(\mathbf{r} \cdot \hat{\mathbf{d}})^2}{r^7} - \frac{6(\mathbf{r} \cdot \hat{\mathbf{d}})d_\alpha}{r^5},$$

the second derivative reads consequently

$$\begin{aligned} \frac{\partial^2 V_{dd}}{\partial r_\alpha \partial r_\beta} &= -3 \frac{\delta_{\alpha\beta}}{r^5} + 15 \frac{r_\alpha r_\beta}{r^7} + 15 \frac{\delta_{\alpha\beta}(\mathbf{r} \cdot \hat{\mathbf{d}})^2}{r^7} - 15 \cdot 7 \frac{r_\alpha r_\beta (\mathbf{r} \cdot \hat{\mathbf{d}})^2}{r^9} \\ &\quad + 30 \frac{d_\alpha r_\beta + d_\beta r_\alpha}{r^7} (\mathbf{r} \cdot \hat{\mathbf{d}}) - 6 \frac{d_\alpha d_\beta}{r^5} \\ &= \delta_{\alpha\beta} \left(-3 \frac{1}{r^5} + 15 \frac{(\mathbf{r} \cdot \hat{\mathbf{d}})^2}{r^7} \right) - 6 \frac{d_\alpha d_\beta}{r^5} + 30 \frac{d_\alpha r_\beta + d_\beta r_\alpha}{r^7} (\mathbf{r} \cdot \hat{\mathbf{d}}) \\ &\quad + 15 r_\alpha r_\beta \left(\frac{1}{r^7} - 7 \frac{(\mathbf{r} \cdot \hat{\mathbf{d}})^2}{r^9} \right). \end{aligned}$$

General Dipole-Dipole Interactions

In the case where not all dipole moments are aligned, the properties of the interaction get extended to

$$V_{dd}(\mathbf{r}) = \frac{\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2}{r^3} - 3 \frac{(\mathbf{r} \cdot \hat{\mathbf{d}}_1)(\mathbf{r} \cdot \hat{\mathbf{d}}_2)}{r^5}.$$

The first derivative reads

$$\frac{\partial V_{dd}}{\partial r_\alpha} = -\frac{3(\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2) r_\alpha}{r^5} + \frac{15r_\alpha(\mathbf{r} \cdot \hat{\mathbf{d}}_1)(\mathbf{r} \cdot \hat{\mathbf{d}}_2)}{r^7} - \frac{3 \left((\mathbf{r} \cdot \hat{\mathbf{d}}_1) d_{2,\alpha} + (\mathbf{r} \cdot \hat{\mathbf{d}}_2) d_{1,\alpha} \right)}{r^5},$$

the second derivative reads consequently

$$\begin{aligned} \frac{\partial^2 V_{dd}}{\partial r_\alpha \partial r_\beta} &= -3 \frac{\delta_{\alpha\beta}(\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2)}{r^5} + 15 \frac{r_\alpha r_\beta (\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2)}{r^7} + 15 \frac{\delta_{\alpha\beta}(\mathbf{r} \cdot \hat{\mathbf{d}}_1)(\mathbf{r} \cdot \hat{\mathbf{d}}_2)}{r^7} - 15 \cdot 7 \frac{r_\alpha r_\beta (\mathbf{r} \cdot \hat{\mathbf{d}}_1)(\mathbf{r} \cdot \hat{\mathbf{d}}_2)}{r^9} \\ &\quad + 15 \left[\frac{d_{1,\alpha} r_\beta + d_{1,\beta} r_\alpha}{r^7} (\mathbf{r} \cdot \hat{\mathbf{d}}_2) + \frac{d_{2,\alpha} r_\beta + d_{2,\beta} r_\alpha}{r^7} (\mathbf{r} \cdot \hat{\mathbf{d}}_1) \right] - 3 \frac{d_{1,\alpha} d_{2,\beta} + d_{2,\alpha} d_{1,\beta}}{r^5} \\ &= \delta_{\alpha\beta} \left(-3 \frac{\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2}{r^5} + 15 \frac{(\mathbf{r} \cdot \hat{\mathbf{d}}_1)(\mathbf{r} \cdot \hat{\mathbf{d}}_2)}{r^7} \right) - 3 \frac{d_{1,\alpha} d_{2,\beta} + d_{2,\alpha} d_{1,\beta}}{r^5} \\ &\quad + 15 \left[\frac{d_{1,\alpha} r_\beta + d_{1,\beta} r_\alpha}{r^7} (\mathbf{r} \cdot \hat{\mathbf{d}}_2) + \frac{d_{2,\alpha} r_\beta + d_{2,\beta} r_\alpha}{r^7} (\mathbf{r} \cdot \hat{\mathbf{d}}_1) \right] \\ &\quad + 15 r_\alpha r_\beta \left(\frac{\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2}{r^7} - 7 \frac{(\mathbf{r} \cdot \hat{\mathbf{d}}_1)(\mathbf{r} \cdot \hat{\mathbf{d}}_2)}{r^9} \right). \end{aligned}$$

For the the Van der Waals interaction the potential has to be squared. Therefore the second derivative reads

$$\begin{aligned} \frac{\partial V_{dd}^2}{\partial r_\alpha} &= 2 V_{dd} \frac{\partial V_{dd}}{\partial r_\alpha} \\ \frac{\partial^2 V_{dd}^2}{\partial r_\alpha \partial r_\beta} &= 2 \frac{\partial V_{dd}}{\partial r_\alpha} \frac{\partial V_{dd}}{\partial r_\beta} + 2 V_{dd} \frac{\partial^2 V_{dd}}{\partial r_\alpha \partial r_\beta} \end{aligned}$$

Winding Number

A chiral Hamiltonian is defined through the property, that it anticommutes with a unitary hermitian operator Γ

$$\Gamma H \Gamma = -H \quad \text{with} \quad \Gamma^2 = 1. \quad (0.1)$$

From this property follows that the Hamiltonian can be written in block off-diagonal form

$$H = \begin{pmatrix} 0 & h \\ h^\dagger & 0 \end{pmatrix}, \quad (0.2)$$

where h is, in general, a non-Hermitian matrix. This structure also applies for the matrix in momentum space and allows the definition of a winding number, which characterizes the topological properties of the system. If the Hamiltonian is 2×2 dimensional, the winding number can be defined through the decomposition of H in terms of the Pauli matrices

$$H = \mathbf{n} \cdot \boldsymbol{\sigma}, \quad (0.3)$$

where $\mathbf{n} = (n_x, n_y, 0)$ is a vector in the xy -plane. The winding number is then given by the winding of the vector $\mathbf{n}$ around the origin when traversing the Brillouin zone. Identifying the xy -plane of $\mathbf{n}$ with the complex plane, which defines a path $\mathbf{n}(k) \rightarrow \gamma(k)$, the winding number can be calculated through

$$w = \frac{1}{2\pi i} \oint_{\gamma_k} \frac{1}{z} dz = \frac{1}{2\pi i} \int_{\text{BZ}} dk \frac{1}{\gamma(k)} \frac{d\gamma(k)}{dk}.$$

In terms of the Hamiltonian above this reads

$$w = \frac{1}{2\pi i} \int_{\text{BZ}} dk \frac{h'(k)}{h(k)}.$$

Generalizing the winding number to higher dimensional Hamiltonians is possible through the use of the block off-diagonal form. The winding number then reads

$$w = \frac{1}{2\pi i} \int_{\text{BZ}} dk \text{Tr} [h^{-1}(k) \partial_k h(k)].$$

Diagonalizing h as $h = S \Lambda S^{-1}$ with the diagonal matrix $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots)$ and the invertible matrix S , the trace can be rewritten as

$$\begin{aligned} \text{Tr} [h^{-1} \partial_k h] &= \text{Tr} [S \Lambda S^{-1} \partial_k (S \Lambda S^{-1})] \\ &= \text{Tr} [\Lambda^{-1} \partial_k \Lambda + S^{-1} \partial_k S + (\partial_k S^{-1}) S] \\ \text{Tr} [S^{-1} \partial_k S + S \partial_k S^{-1}] &= \text{Tr} [\partial_k (S S^{-1})] = 0, \\ \Rightarrow \text{Tr} [h^{-1} \partial_k h] &= \text{Tr} [\Lambda^{-1} \partial_k \Lambda] = \sum_i \frac{\partial_k \lambda_i}{\lambda_i} = \partial_k \left(\sum_i \ln(\lambda_i) \right) = \partial_k \ln(\det[h]). \end{aligned}$$

Another useful definition of the winding number is through the so-called Q -matrix, which is defined through the projectors onto the eigenstates of the Hamiltonian. As the Hamiltonian is chiral, the eigenvalues come in pairs $\pm E$. We introduce a labeling of the eigenvalues, such that $E_j = -E_{-j}$ with $j = 1, 2, \dots, N$ and N being half the dimension of the Hamiltonian. The corresponding eigenstates are labeled accordingly ($|\psi_j\rangle$) and posses the property $\Gamma |\psi_j\rangle = |\psi_{-j}\rangle$. The Q -matrix is defined as

$$Q = \sum_{j=1}^N (|\psi_j\rangle \langle \psi_j| - |\psi_{-j}\rangle \langle \psi_{-j}|).$$

As a consequence of the chiral symmetry, the Q -matrix anticommutes with Γ and can therefore be written in block off-diagonal form in the basis, where Γ is diagonal, which is also the case in which H is

block off-diagonal.

$$\begin{aligned}\Gamma Q \Gamma &= \sum_{j=1}^N (|\psi_{-j}\rangle \langle \psi_{-j}| - |\psi_j\rangle \langle \psi_j|) = -Q \\ \Rightarrow Q &= \begin{pmatrix} 0 & q \\ q^\dagger & 0 \end{pmatrix}\end{aligned}$$

Labeling the eigenvalues and eigenstates of h as h_i and $|h_i\rangle$ respectively, a basis of H can be constructed that preserves the block off-diagonal structure. The basis is split into two sets A and B,

$$|h_i^A\rangle = \begin{pmatrix} |h_i\rangle \\ 0 \end{pmatrix}, \quad |h_i^B\rangle = \begin{pmatrix} 0 \\ |h_i\rangle \end{pmatrix}.$$

The eigenstates of H can be constructed from this as

$$\begin{aligned}|H_i^+\rangle &= \frac{1}{\sqrt{2}|h_i|} (|h_i\rangle |h_i^A\rangle + h_i^* |h_i^B\rangle) \\ |H_i^-\rangle &= \frac{1}{\sqrt{2}|h_i|} (|h_i\rangle |h_i^A\rangle - h_i^* |h_i^B\rangle) \\ \text{with } H |H_i^\pm\rangle &= \pm |h_i| |H_i^\pm\rangle.\end{aligned}$$

The Q -matrix can thus be written as

$$\begin{aligned}Q &= \sum_i (|H_i^+\rangle \langle H_i^+| - |H_i^-\rangle \langle H_i^-|) \\ &= \sum_i \frac{1}{|h_i|} (h_i |h_i^A\rangle \langle h_i^B| + h_i^* |h_i^B\rangle \langle h_i^A|)\end{aligned}$$

It now follows that the winding number of the q -matrix is the same as that of h

$$\begin{aligned}w_q &= \frac{1}{2\pi i} \sum_j \int_{\text{BZ}} dk \frac{\partial_k \hat{h}_j}{|\hat{h}_j|} \\ &= \frac{1}{2\pi i} \sum_j \int_{\text{BZ}} dk \frac{\partial_k h_j}{h_j} \\ &= w_h.\end{aligned}$$

where $\hat{h}_j = h_j/|h_j|$ is the normalized eigenvalue. Another equivalent expression for the winding number can be given through the eigenstates of the Hamiltonian

$$\begin{aligned}w &= \frac{1}{\pi i} \sum_j \int_{\text{BZ}} dk S_j(k) \\ \text{with } S_j(k) &= \langle \Gamma H_j^- | \partial_k | H_j^- \rangle \\ &= \langle H_j^+ | \partial_k | H_j^- \rangle \\ &= \frac{1}{2} (\langle h_j^A | \partial_k | h_j^A \rangle - \langle h_j^B | \partial_k | h_j^B \rangle) - \frac{1}{2} \frac{h_j \partial_k h_j^*}{|h_j|^2} \\ &= -\frac{1}{2} \frac{\partial_k h_j^*}{h_j^*}.\end{aligned}$$